L15

1 How many numbers are there smaller than 1000 that have exactly 10 factors?

1. 【Answer】 22
【Solution】 $2^9 = 512$;
$2^4 \times 3$、$2^4 \times 5$、$2^4 \times 7$、$2^4 \times 11$、$2^4 \times 13$、$2^4 \times 17$、$2^4 \times 19$、$2^4 \times 23$、$2^4 \times 29$、$2^4 \times 31$、$2^4 \times 37$、$2^4 \times 41$、$2^4 \times 43$、$2^4 \times 47$、$2^4 \times 53$、$2^4 \times 59$、$2^4 \times 61 = 976$; $3^4 \times 2$、$3^4 \times 5$、$3^4 \times 7$、$3^4 \times 11 = 891$

5 Divide the following eight numbers into 2 groups so that the 4 numbers in each group have an equal product.

15, 19, 30, 39, 46, 57, 92, 117.

5. 【Answer】 There is more than one way.
(19, 30, 46, 117) and (15, 39, 57, 92)
(19, 15, 92, 117) and (30, 39, 57, 46) .
【Solution】 $15 = 1 \times 15$
$30 = 2 \times 15$
$46 = 2 \times 23$
$19 = 1 \times 19$
$39 = 3 \times 13$
$57 = 3 \times 19$
$92 = 2 \times 2 \times 23$
$117 = 3 \times 3 \times 13$
Each group needs $15 \times 19 \times 23 \times 13$, so we can get:
(19, 30, 46, 117) and (15, 39, 57, 92)
(19, 15, 92, 117) and (30, 39, 57, 46) .

7 Divide 14, 21, 22, 26, 33, 35, 39, 55, and 65 into 3 groups so that the 3 numbers in each group have an equal product.

7. 【Answer】 There is more than one answer:
(14, 39, 55), (21, 22, 65), (26, 33, 35).
【Solution】 $14 = 2 \times 7$, $26 = 2 \times 13$, $39 = 3 \times 13$,
$21 = 3 \times 7$, $33 = 3 \times 11$, $55 = 5 \times 11$,
$22 = 2 \times 11$, $35 = 5 \times 7$, $65 = 13 \times 5$,
The analysis above shows that it is divided into three groups, and the product of each group is $7 \times 11 \times 13 \times 2 \times 3 \times 5$,
①$7 \times 11 \times 13 \times 2 \times 3 \times 5 = 14 \times 39 \times 55$;
②$7 \times 11 \times 13 \times 2 \times 3 \times 5 = 21 \times 22 \times 65$;
③$7 \times 11 \times 13 \times 2 \times 3 \times 5 = 26 \times 33 \times 35$.
The three groups of numbers are: (14, 39, 55), (21, 22, 65), (26, 33, 35).

9 How many consecutive 0s are at the end of the product of the 100 natural numbers from 101 to 200?

9. 【Answer】 25
【Solution】 Among $1 \times 2 \times 3 \cdots \times 200$, there are $200 \div 5 + 40 \div 5 + [8 \div 5] = 49$ zeros at the end.
Among $1 \times 2 \times 3 \cdots \times 100$, there are $100 \div 5 + 20 \div 5 = 24$ zeros at the end.
$49 - 24 = 25$

1 A natural number has 16 factors. What is the minimum value of this natural number?

1. 【Answer】 120
【Solution】 Only 1 prime factor: $16 - 1 = 15$, $N = p^{15}$,
Only 2 prime factors: $16 = 2 \times 8 = 4 \times 4$, $N = p^7 \times q^1 = p^3 \times q^3$,
Only 3 prime factors: $16 = 2 \times 2 \times 4$, $N = p^3 \times q \times r$,
Only 4 prime factors: $16 = 2 \times 2 \times 2 \times 2$, $N = p \times q \times r \times s$
Thus, the smallest one is 120.

2 The natural numbers a and b each have exactly 5 different factors. How many different factors does ab have?

2. 【Answer】 9 or 25
【Solution】 A natural number has exactly 5 factors, then it must be a prime number to the power of 4.
If $a = b = p^4$, then $ab = p^8$ has 9 factors.
If $a \neq b$, let $a = p^4$ and $b = q^4$, then ab has $(4 + 1) \times (4 + 1) = 25$ factors.

4 a has exactly 9 factors. How many factors does a^3 have at most?

4. 【Answer】 49
【Solution】 According to the theorem of number of factors, a may be written as p^8, in which case $a^3 = p^{24}$, with 25 factors.
But 9 is a composite number, $9 = 3 \times 3$, and $a = p^2 \times q^2$.
This time, a has exactly $(2 + 1) \times (2 + 1) = 9$ factors.
Thus, $a^3 = p^6 \times q^6$, with $(6 + 1) \times (6 + 1) = 49$ factors.
a^3 has a maximum of 49 factors.

10 There are 2 two-digit natural numbers whose greatest common factor is 6 and least common multiple is 90. The sum of these two numbers is _____ .

A. 96 B. 60 C. 48 D. 30

10. 【Answer】 C
【Solution】 Let these two numbers be $6a$ and $6b$. Therefore, $ab = 90 \div 6 = 15$. So, a and b are 3 and 5, respectively. Therefore, the sum of these two numbers is $6 \times (3 + 5) = 48$.

L16

7 Given that $7x \equiv 107 \pmod{11}$, the minimum value of $|x|$ is _____ .

7. 【Answer】 2
【Solution】 $7x \equiv 107 \pmod{11}$,
$7x \equiv 107 - 44 \pmod{11}$,
$7x \equiv 63 \pmod{11}$,
$x \equiv 9 \pmod{11}$,
$\therefore x = -2, |x| = 2$.

5 Prove that $3^{2012} + 4^{2013}$ is divisible by 5.

5. 【Answer】 See solution.
【Solution】 $3^{2012} + 4^{2013} \equiv 9^{1006} + 4^{2013} \equiv (-1)^{1006} + (-1)^{2013} \equiv 1 + (-1) = 0 \pmod{5}$.

7 There exists an integer greater than 10, such that 39, 51, and 147 have the same remainder when being divided by this integer. Find this integer.

7. 【Answer】 12 .
【Solution】 Let the integer be x, then 39 , 51 , 147 divided by x have the same remainder. Therefore, the difference between any two of these numbers is divisible by x . $51 - 39 = 12$, $147 - 51 = 96$. Becuase $(12, 96) = 12$, and this integer is greater than 10, we have $x = 12$.

8 What is the remainder when $1^1 + 2^2 + 3^3 + 4^4 + 5^5 + 6^6 + 7^7 + 8^8 + 9^9$ is divided by 3?

8. 【Answer】 1
【Solution】 First we observe that:
$1 \equiv 4 \equiv 7 \equiv 1 \pmod 3$
$2 \equiv 5 \equiv 8 \equiv -1 \pmod 3$ $3 \equiv 6 \equiv 9 \equiv 0 \pmod 3$
Therefore, the sum can be simplified with modulo congruence:
$1^1 + 2^2 + 3^3 + 4^4 + 5^5 + 6^6 + 7^7 + 8^8 + 9^9 \equiv 1^1 + (-1)^2 + 0 + 1^4 + (-1)^5 + 0 + 1^7 + (-1)^8 + 0 \equiv 4 \equiv 1 \pmod 3$.
The remainder is then 3.

9 Given a number greater than 1, when divided by 298, 243, and 208, the remainders are a, $a+2$, and $a+5$, respectively. What is the value of this natural number?

9. 【Answer】 19

【Solution】 This number is congruent to 298, 241, and 203. Therefore, it is divisible by 57, 38, and 95, and the number itself is greater than 1. Hence, the number is 19.

L17

7 There are segments of length 2, 3, 4, 5, 6, 7, 8, and 9, and three of them are selected to form a triangle, so how many different options are there?

1. 【Answer】 34.
【Solution】 The sum of the two sides of the triangle is greater than the third side, which is classified as follows:
(2,3,4), (2,4,5), (2,5,6), (2,6,7), (2,7,8), (2,8,9);
(3,4,5), (3,4,6), (3,5,6), (3,5,7), (3,6,7), (3,6,8), (3,7,8), (3,7,9), (3,8,9);
(4,5,6) , (4,5,7) , (4,5,8) , (4,6,7), (4,6,8), (4,6,9), (4,7,8), (4,7,9), (4,8,9);
(5,6,7), (5,6,8), (5,6,9), (5,7,8), (5,7,9), (5,8,9);
(6,7,8), (6,7,9), (6,8,9);
(7,8,9).
Total: $6+9+9+6+3+1=34$.

3 The teacher distributes 4 chocolate bars each in a different flavor to 4 of a group of 8 kids. There are _____ ways to distribute the chocolate bars.

3. 【Answer】 1680
【Solution】 Because each chocolate bar is different, this is a permutation problem to pick 4 kids among 8. Thus, ${}_8P_4 = 1680$.

4 How many kinds of triangles are there with lengths all integers and less than 5 on three sides? (As long as the triangles are with the same data of the length of three sides, they are the same.) _____

4. 【Answer】 13
【Solution】 isosceles: (111)(221)(222)(223)(331)(332)(333)(334)(441)(442)(443)(444), 12 ways;
non-isosceles: (234), 1 way.
So 13 in total.

8 a, b, c are integers, and $a+b+c=12$. How many triple-digit numbers are there that meet this condition? (a, b, c can be the same)

A. 52 B. 53 C. 54 D. 55

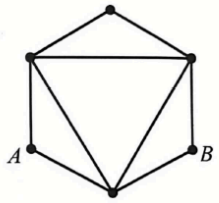
8. 【Answer】 A
【Solution】 Insertion method, $C_{11}^{2}=55$, $a=10, b=1, c=1; a=1, b=10, c=1; a=1, b=1, c=10$ cannot be the answer, so there are $55-3=52$ numbers.

1 A team has 5 boys, 4 girls. Now we need to pick 3 people from this team to do the cleaning.

(3) If exactly one of A, B is picked, there are _____ ways of picking.

1. 【Answer】 (1) 40
(2) 28
(3) 42
【Solution】 (1) $C_5^2 \times C_4^1 = 10 \times 4 = 40$.
(2) $1 \times C_8^2 = 28$.
(3) $C_2^1 \times C_7^2 = 2 \times 21 = 42$.

4 As shown in the picture, start from the point A, walk along the line sections, and finally reach the point B.

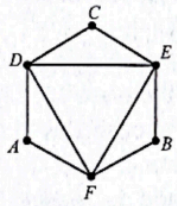


(1) If pass each black dot once, how many ways to go?

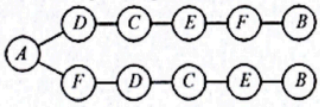
(2) If pass each black dot at least once, how many ways to go?

4. 【Answer】 (1) 2.
(2) 10.

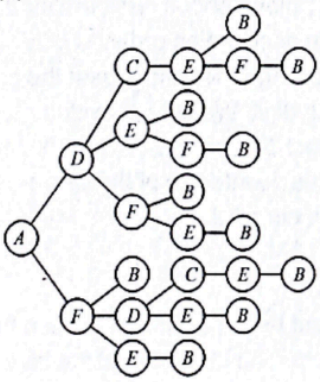
【Solution】 (1) Letter the other dots:



So 2 ways to go are:



(2) The 10 ways are shown in the following picture:



5 There is a batch of thin wooden strips with lengths of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 and 11 cm, and they are numerous enough to select 3 as three sides to enclose a triangle. If one of the sides is specified to be 11 cm long, how many different triangles can you have? _____

5. 【Answer】 36

【Solution】 Just make the sum of the lengths of the other two sides greater than 11. These two sides may be: (11, 11); (10, 10); (10, 11); (9, 9); (9, 10); (9, 11); (8, 8); (8, 9); (8, 10); (8, 11); (7, 7); (7, 8); (7, 9); (7, 10); (7, 11); (6, 6); (6, 7); (6, 8); (6, 9); (6, 10); (6, 11); (5, 7); (5, 8); (5, 9); (5, 10); (5, 11); (4, 8); (4, 9); (4, 10); (4, 11); (3, 9); (3, 10); (3, 11); (2, 10); (2, 11); (1, 11).

It can be enclosed in 36 different triangles.

6 Call a positive integer monotonous if it is a one–digit number or its digits, when read from left to right, form either a strictly increasing or a strictly decreasing sequence. For example,3,23578, and 987620 are monotonous, but 88,7434, and 23557 are not. How many monotonous positive integers are there? (2017 AMC 10B Problems, Question #17)

A. 1024 B. 1524 C. 1533 D. 1536 E. 2048

6. 【Answer】 B

【Solution】 Case 1: monotonous numbers with digits in ascending order.

There are $\sum_{n=1}^{9}\binom{9}{n}$ ways to choose n digits from the digits 1 to 9. For each of these ways, we can generate exactly one monotonous number by ordering the chosen digits in ascending order. Note that 0 is not included since it will always be a leading digit and that is not allowed. Also, ∅(the empty set) isn't included because it doesn't generate a number. The sum is equivalent to $\sum_{n=1}^{9}\binom{9}{n}-\binom{9}{0}=2^9-1=511.$

Case 2: monotonous numbers with digits in descending order.

There are $\sum_{n=0}^{10}\binom{10}{n}$ ways to choose n digits from the digits 0 to 9. For each of these ways, we can generate exactly one monotonous number by ordering the chosen digits in descending order. Note that 0 is included since we are allowed to end numbers with zeros. However,∅ (the empty set) still isn't included because it doesn't generate a number. The sum is equivalent to $\sum_{n=0}^{10}\binom{10}{n}-\binom{10}{0}=2^{10}-1=1023.$

We discard the number 0 since it is not positive. Thus there are 1022 here.

Since the 1–digit numbers 1 to 9 satisfy both case 1 and case 2, we have overcounted by 9. Thus there are $511+1022-9=1524$ monotonous numbers.

Like Solution 1, divide the problem into an increasing and decreasing case:

Case 1: Monotonous numbers with digits in ascending order.

Arrange the digits 1 through 9 in increasing order, and exclude 0 because a positive integer cannot begin with 0.

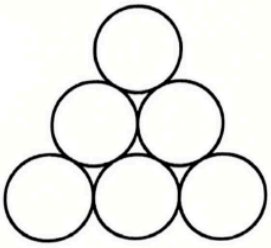
To get a monotonous number, we can either include or exclude each of the remaining 9 digits, and there are $2^9=512$ ways to do this. However, we cannot exclude every digit at once, so we subtract 1 to get $512-1=511$ monotonous numbers for this case.

Case 2: Monotonous numbers with digits in descending order.

This time, we arrange all 10 digits in decreasing order and repeat the process to find $2^{10}=1024$ ways to include or exclude each digit. We cannot exclude every digit at once, and we cannot include only 0, so we subtract 2 to get $1024-2=1022$ monotonous numbers for this case. At this point, we have counted all of the single-digit monotonous numbers twice, so we must subtract 9 from our total.

Thus our final answer is $511+1022-9=1524$.

7 In the figure below, 3 of the 6 disks are to be painted blue, 2 are to be painted red, and 1 is to be painted green. Two paintings that can be obtained from one another by a rotation or a reflection of the entire figure are considered the same. How many different paintings are possible? ( ) (2017 AMC 12B Problem, Question #13)



A. 6 B. 8 C. 9 D. 12 E. 15

7. 【Answer】 D

【Solution】 First we figure out the number of ways to put the blue 3 disks. Denote the spots to put the disks as $1-6$ from left to right, top to bottom. The cases to put the blue disks are $(1,2,3)$, $(1,2,4),(1,2,5),(1,2,6),(2,3,5),(1,4,6)$. For each of those cases we can easily figure out the number of ways for each case, so the total amount is $2+2+3+3+1+1=12$.

L18

1 A bag contains 4 balls that are identical in size and texture, including 2 red balls and 2 green balls. Two balls are randomly drawn one after another without replacement. What is the probability that the two balls are of the same color? ( ) .

A. $\frac{1}{4}$ B. $\frac{1}{3}$ C. $\frac{1}{2}$ D. $\frac{2}{3}$

1. 【Answer】 B
【Solution】 The total number of basic outcomes is
$n = {}_4C_2 = 6$.
There are 2 outcomes where the two balls are the same color.
Therefore, the probability that the two balls are the same color is
$\frac{2}{6} = \frac{1}{3}$.
Thus, the correct answer is: B.

3 If one number is chosen at random from all the positive divisors of 2025, what is the probability that this number is a perfect square?

A. 0.4 B. 0.3 C. 0.6 D. 0.7 E. 0.8

3. 【Answer】 A
【Solution】 Since $2025 = 3^4 \times 5^2$,
the positive divisors of 2025 are
$1, 3, 5, 9, 15, 25, 27, 45, 75, 81, 135, 225, 405, 675, 2025$,
a total of 15 numbers.
Among them, the perfect squares are
$1, 9, 25, 81, 225, 2025$,
a total of 6 numbers.
Therefore, if one number is chosen at random from all the positive divisors of 2025, the probability that it is a perfect square is
$\frac{6}{15} = \frac{2}{5}$.
Hence, the answer is $\frac{2}{5}$.

4 Randomly choose an integer from 1 to 6, the probability distribution follows:

$P(6) = P(1)$;

$P(2) = P(4)$;

$P(3) = 3P(5)$;

$P(6) = 2P(4) = 4P(5)$.

Find $P(1)$.

4. 【Answer】 $\frac{1}{4}$

【Solution】 Let p denote the probabilty that a five is selected. Then we can see that $P(2) = P(4) = 2p$ and $P(1) = P(6) = 4p$, $P(3) = 3p$. Since all of these probabilities must add to one, we get $1 = P(1) + P(2) + P(3) + P(4) + P(5) + P(6) = 16p$. Therefore, $p = \frac{1}{16}$ and $P(1) = \frac{1}{4}$.

5 Assign volunteers A, B, C to classrooms numbered $1, 2, 3$ for cleaning, with exactly one volunteer in each classroom. What is the probability that A is not assigned to classroom 3?

A. $\frac{2}{3}$ B. $\frac{3}{4}$ C. $\frac{1}{4}$ D. $\frac{1}{3}$ E. $\frac{3}{5}$

5. 【Answer】 A

【Solution】 Volunteers A, B, C are assigned to classrooms numbered $1, 2, 3$,
with exactly one volunteer in each classroom. The possible assignments are:
$(A, 1), (B, 2), (C, 3)$,
$(A, 1), (B, 3), (C, 2)$,
$(A, 2), (B, 1), (C, 3)$,
$(A, 2), (B, 3), (C, 1)$,
$(A, 3), (B, 1), (C, 2)$,
$(A, 3), (B, 2), (C, 1)$,
a total of 6.
Among these, the cases where A is not assigned to classroom 3 are:
$(A, 1), (B, 2), (C, 3)$,
$(A, 1), (B, 3), (C, 2)$,
$(A, 2), (B, 1), (C, 3)$,
$(A, 2), (B, 3), (C, 1)$,
a total of 4.
Therefore, the probability that A is not assigned to classroom 3 is
$P = \frac{4}{6} = \frac{2}{3}$.
Thus, the answer is A.

6 To assess students' physical fitness, 3 items are chosen at random from a total of 7 events: 3 track and field events, 2 ball games, and 2 martial arts events. What is the probability that the chosen 3 items come from exactly two categories?

A. $\frac{22}{35}$ B. $\frac{2}{7}$ C. $\frac{4}{7}$ D. $\frac{11}{35}$ E. $\frac{33}{35}$

6. 【Answer】 A

【Solution】 The total number of ways to choose 3 items from these 7 events is ${}_7C_3 = 35$.

The number of ways where all 3 items come from the same category is ${}_3C_3 = 1$.

The number of ways where the 3 items cover all three categories is ${}_3C_1\,{}_2C_1\,{}_2C_1 = 12$.

Therefore, the number of favorable cases (exactly two categories) is $35 - 1 - 12 = 22$.

So the required probability is $\frac{22}{35}$.

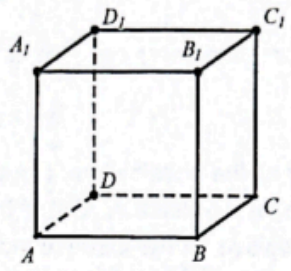
Hence, the answer is $\frac{22}{35}$.

7 If three vertices are chosen at random from the eight vertices of a cube and connected to form a triangle, what is the probability that the triangle is equilateral?

A. $\frac{1}{7}$ B. $\frac{1}{14}$ C. $\frac{2}{7}$ D. $\frac{4}{35}$ E. $\frac{5}{14}$

1. 【Answer】 A

【Solution】 The number of ways to choose any three vertices from the eight vertices to form a triangle is ${}_8C_3 = 56$.



Among them, there are 8 outcomes that form an equilateral triangle:

$\triangle ACD_1$, $\triangle BDC_1$, $\triangle ACB_1$, $\triangle BDA_1$, $\triangle A_1C_1B$, $\triangle B_1D_1A$, $\triangle B_1D_1C$, $\triangle A_1C_1D$.

Therefore, the probability is $\frac{8}{56} = \frac{1}{7}$, hence the answer is A.

2 One fair die has faces 1, 1, 2, 2, 3, 3 and another has faces 4, 4, 5, 5, 6, 6. The dice are rolled and the numbers on the top faces are added. What is the probability that the sum will be odd?

(2005 AMC 10B Problem, Question #9)

A $\frac{1}{3}$ B. $\frac{4}{9}$ C. $\frac{1}{2}$ D. $\frac{5}{9}$ E. $\frac{2}{3}$

2. 【Answer】 D

【Solution】 In order to obtain an odd sum, exactly one out of the two dice must have an odd number. We can easily find the total probability using casework.

Case 1: The first die is odd and the second die is even.

The probability of this happening is $\frac{4}{6} \times \frac{4}{6} = \frac{16}{36} = \frac{4}{9}$.

Case 2: The first die is even and the second die is odd.

The probability of this happening is $\frac{2}{6} \times \frac{2}{6} = \frac{4}{36} = \frac{1}{9}$.

Adding these two probabilities will give us our final answer. $\frac{4}{9} + \frac{1}{9} = \frac{5}{9}$.

6 Suppose that one of every 500 people in a certain population has a particular disease, which displays no symptoms. A blood test is available for screening for this disease. For a person who has this disease, the test always turns out positive. For a person who does not have the disease, however, there is a 2% false positive rate - in other words, for such people, 98% of the time the test will turn out negative, but 2% of the time the test will turn out positive and will incorrectly indicate that the person has the disease. Let p be the probability that a person who is chosen at random from this population and gets a positive test result actually has the disease. Which of the following is closest to p?

(2012 AMC 10B Problem, Question #18)

A. $\frac{1}{98}$ B. $\frac{1}{9}$ C. $\frac{1}{11}$ D. $\frac{49}{99}$ E. $\frac{98}{99}$

6. 【Answer】 C

【Solution】 This question can be solved by considering all the possibilities: 1 out of 500 people will have the disease and will be tested positive for the disease. Out of the remaining 499 people, 2%, or 9.98 people, will be tested positive for the disease incorrectly. Therefore, p can be found by taking $\frac{1}{1+9.98}$, which is closest to $\frac{1}{11}$, (C).

7 A box contains three bags. One bag contains one white pebble and three black pebbles. Another bag also contains one white pebble and three black pebbles. The remaining bag contains one white pebble and four black pebbles. Eddie reaches into the box and randomly pulls out a bag, then reaches into the bag and randomly pulls out a pebble. What is the probability that this pebble is white?

A. $\frac{7}{30}$ B. $\frac{3}{13}$ C. $\frac{1}{9}$ D. $\frac{1}{4}$ E. $\frac{11}{36}$

7. 【Answer】 A

【Solution】 $\frac{1}{3}\times\frac{1}{4}+\frac{1}{3}\times\frac{1}{4}+\frac{1}{3}\times\frac{1}{5}=\frac{7}{30}$.

The answer is A .

8. Forty slips are placed into a hat, each bearing a number 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10, with each number entered on four slips. Four slips are drawn from the hat at random and without replacement. Let p be the probability that all four slips bear the same number. Let q be the probability that two of the slips bear a number a and the other two bear a number $b \neq a$. What is the value of $\frac{q}{p}$?

(2005 AMC 10B Problem, Question #21)

A. 162 B. 180 C. 324 D. 360 E. 720

8. 【Answer】 A

【Solution】 There are 10 ways to determine which number to pick. There are 4! ways to then draw those four slips with that number, and $40 \cdot 39 \cdot 38 \cdot 37$ total ways to draw four slips. Thus $p = \frac{10 \cdot 4!}{40 \cdot 39 \cdot 38 \cdot 37}$.

There are $\binom{10}{2} = 45$ ways to determine which two numbers to pick for the second probability. There are $\binom{4}{2} = 6$ ways to arrange the order which we draw the non-equal slips, and in each order there are $4 \times 3 \times 4 \times 3$ ways to pick the slips, so $q = \frac{45 \cdot 6 \cdot 4^2 \cdot 3^2}{40 \cdot 39 \cdot 38 \cdot 37}$.

Hence, the answer is $\frac{q}{p} = \frac{45 \cdot 6 \cdot 4^2 \cdot 3^2}{10 \cdot 4!} = 162$.

For probability p, there are $\binom{10}{1} = 10$ ways to choose the number you want to show up 4 times.

Hence, the probability is $\frac{10}{\binom{40}{4}}$.

For probability q, there are $\binom{10}{2} = 45$ ways to choose the 2 numbers you want to show up twice.

There are $\binom{4}{2} \cdot \binom{4}{2}$ to pick which slips you want out of the 4 of each.

Hence, the probability is $\frac{45 \cdot 6 \cdot 6}{\binom{40}{4}}$.

Hence, $\frac{q}{p} = \frac{45 \cdot 6 \cdot 6}{10} = 9 \cdot 6 \cdot 3 = 162 \Rightarrow$ A.